

EduGenie – Google Gemini Powered Learning Assistant

Performance Testing

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Testing Approach

Testing focused on functional correctness, validation behavior, resilience to missing credentials/network, model fallback, quiz generation, learning-path generation, and long-input summarization.

Test ID	Scenario	Expected Result	Status
TC-01	Valid academic Q&A	AI-generated answer returned.	Pass
TC-02	Empty question	Validation error; no unnecessary AI call.	Pass
TC-03	Missing GEMINI_API_KEY	Application starts in offline/demo mode.	Pass
TC-04	No internet during explanation	Graceful fallback/offline response.	Pass
TC-05	3-question quiz	Questions, options and answer key returned.	Pass
TC-06	Primary model unavailable/rate-limited	Alternative model selected automatically.	Pass
TC-07	Learning path request	Ordered, leveled roadmap returned.	Pass
TC-08	Very long summary input	Concise summary generated without application failure.	Pass

Performance Acceptance

Performance is primarily dependent on network and Gemini API latency. The architecture minimizes local processing and uses a shared service layer; future production deployment should add measured latency targets, concurrency testing, rate-limit handling, and observability.